Mini Project 1: Caltech-101 Image Classification

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1 Introduction

Image classification has progressed from classical machine learning methods based on hand-crafted features to deep learning models that learn hierarchical representations directly from data. Understanding the trade-offs among model architectures, input resolutions, and training strategies remains important for both research and practical applications.

In this project, I compare three approaches on the Caltech-101 dataset: HOG + SVM as a classical baseline, and two pretrained deep learning models, EfficientNet-B0 and ResNet18.

The results show that (1) deep learning methods substantially outperform the classical approach; (2) EfficientNet-B0 achieves higher accuracy than ResNet18 despite having fewer parameters, highlighting architectural efficiency; (3) increasing input resolution improves performance; and (4) data augmentation does not always lead to gains in transfer learning.

Code and visualizations are available at `https://github.com/Sophie374/caltech101_classification`.

2 Methods

2.1 Dataset

Caltech-101 [1] contains 102 categories (101 object classes plus one background class) with 9,144 images in total. The dataset is moderately imbalanced, with an average of 89.6 images per class and a range from 5 to over 100 images.

This project formulates the task as a 102-class classification problem by including the background category to better reflect real-world "none-of-the-above" scenarios.

2.2 Data Preprocessing

A stratified train/validation/test split (70%/15%/15%) was used to maintain class balance across splits. For deep learning models, images were resized to 224×224 and normalized using ImageNet statistics [2]. Training data was

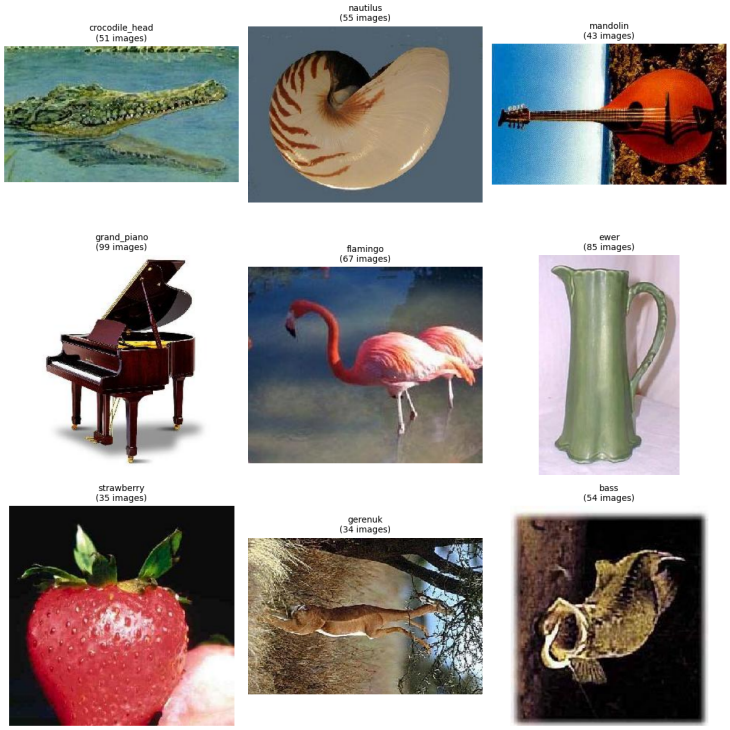

Figure 1: Sample images from the Caltech-101 dataset.

augmented with random resized crop (1.14× scale), horizontal flip (p=0.5), rotation (±15°), and color jitter. Validation and test sets used only resizing and normalization. For HOG, images were resized to 64×64.

2.3 Model Architectures

Three approaches were evaluated:

- **HOG + SVM:** HOG features [3] were extracted from 64×64 images using 9 orientation bins, 8×8 cells, and 2×2 blocks, producing 1,764-dimensional vectors. Features were standardized and used to train a linear SVM ($C = 1.0$).
- **EfficientNet-B0:** A pretrained EfficientNet-B0 [4] (4.1M parameters) was fine-tuned by replacing the classifier with dropout (0.2) followed by a linear layer mapping to 102 output classes. All parameters were trainable.
- **ResNet18:** A pretrained ResNet18 [5] (11.2M parameters) was fine-tuned by replacing the final fully connected layer with a linear layer mapping 512 features to 102 output classes. All parameters were trainable.

2.4 Training and Evaluation

Models were trained in PyTorch on a Tesla T4 GPU using Adam (lr = 10^{-4}), batch size 32, cross-entropy loss, and 10 epochs. Random seed 42 was used for reproducibility, and the best checkpoint was selected based on validation accuracy.

Evaluation metrics included overall accuracy, per-class accuracy, top-5 accuracy, macro and weighted precision/recall/F1-scores, and confusion matrices.

2.5 Ablation Studies

Two ablation studies were conducted using EfficientNet-B0:

- **Image Size:** Compared 64×64 and 128×128 inputs with other hyperparameters fixed.
- **Data Augmentation:** Compared training with and without augmentation using 224×224 inputs.

All ablations used the same training and evaluation protocol as the main experiments.

3 Results

3.1 Model Comparison

Table 1 summarizes test performance across all models. EfficientNet-B0 achieved the highest overall accuracy (94.97%), followed by ResNet18 (93.22%) and HOG + SVM (66.47%). Deep learning models substantially outperformed the classical baseline across all metrics.

Table 1: Comprehensive Model Performance Metrics on Caltech-101 Test Set

Model	**Accuracy**	**Top-5**	**Macro Average**			**Weighted Average**		
		Accuracy	**Precision**	**Recall**	**F1-Score**	**Precision**	**Recall**	**F1-Score**
HOG + SVM	66.47%	82.65%	59.97%	51.58%	53.51%	68.38%	66.47%	65.93%
EfficientNet-B0	**94.97%**	**99.49%**	**93.20%**	**91.94%**	**92.11%**	**95.21%**	**94.97%**	**94.83%**
ResNet18	93.22%	99.42%	92.47%	89.29%	89.93%	94.00%	93.22%	93.06%

3.2 Individual Model Results

3.2.1 HOG + SVM

HOG + SVM achieved 66.47% accuracy and 82.65% top-5 accuracy. Macro precision, recall, and F1-score were 59.97%, 51.58%, and 53.51%, respectively. Several underrepresented classes (e.g., brontosaurus, flamingo_head, lotus) had 0% accuracy, while a few categories achieved perfect classification. Figure 2 presents the confusion matrix and per-class accuracy analysis.

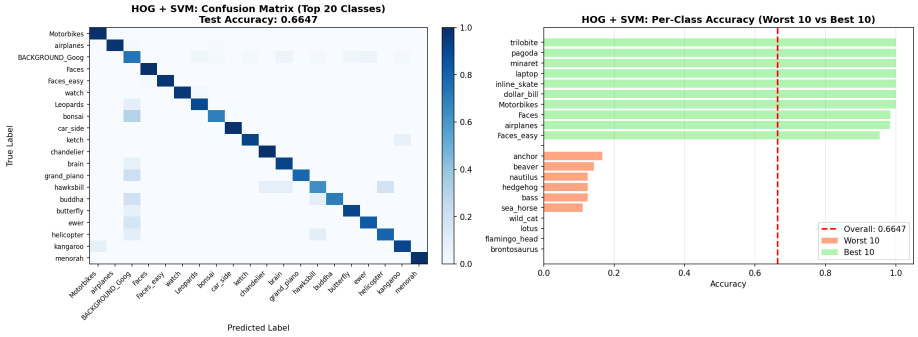

Figure 2: HOG + SVM model performance: (left) confusion matrix for top 20 classes, (right) per-class accuracy showing worst 10 and best 10 classes.

3.2.2 EfficientNet-B0

EfficientNet-B0 achieved 94.97% accuracy with 99.49% top-5 accuracy. Macro precision, recall, and F1-score were 93.20%, 91.94%, and 92.11%. The lowest per-class accuracy was 16.67% (lobster, 6 test samples), while many classes achieved perfect predictions. Training curves and detailed visualizations are shown in Figure 3.

3.2.3 ResNet18

ResNet18 achieved 93.22% accuracy and 99.42% top-5 accuracy, with macro precision, recall, and F1-score of 92.47%, 89.29%, and 89.93%. Certain classes (e.g., water_lilly, lobster) showed lower performance. Visualizations are provided in Figure 4.

3.3 Ablation Studies

3.3.1 Image Size Ablation

Increasing input resolution from 64×64 to 128×128 improved accuracy from 81.63% to 94.17% (+12.54 percentage points), with consistent gains across macro precision, recall, and F1-score (Table 2). Figure 5 shows the corresponding training dynamics and metric comparisons.

Table 2: Image Size Ablation Study Results

Image Size	**Accuracy**	**Precision (Macro)**	**Recall (Macro)**	**F1-Score (Macro)**
64×64	81.63%	75.24%	72.77%	72.52%
128×128	**94.17%**	**92.25%**	**91.27%**	**90.83%**

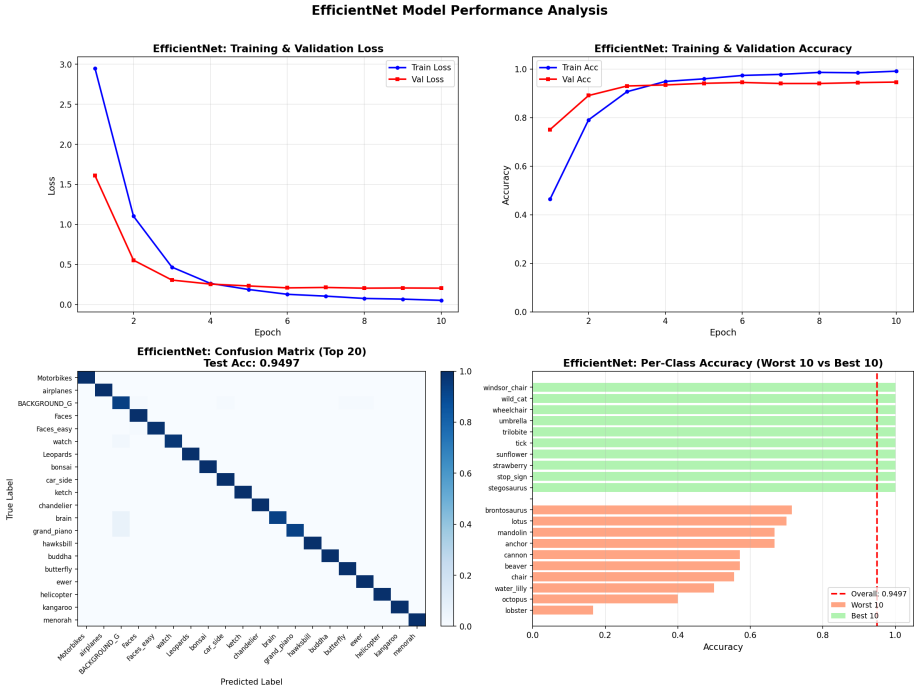

Figure 3: EfficientNet-B0 performance: (top) training and validation loss/accuracy curves, (bottom left) confusion matrix for top 20 classes, (bottom right) per-class accuracy.

3.3.2 Data Augmentation Ablation

Table 3 compares training with and without augmentation using 224×224 inputs. Removing augmentation improved accuracy from 94.97% to 96.57% (+1.60 percentage points). Detailed training curves and metric comparisons are shown in Figure 6.

Table 3: Data Augmentation Ablation Study Results

Configuration	**Accuracy**	**Precision (Macro)**	**Recall (Macro)**	**F1-Score (Macro)**
With Augmentation	94.97%	93.20%	91.94%	92.11%
Without Augmentation	**96.57%**	**95.88%**	**94.91%**	**94.94%**

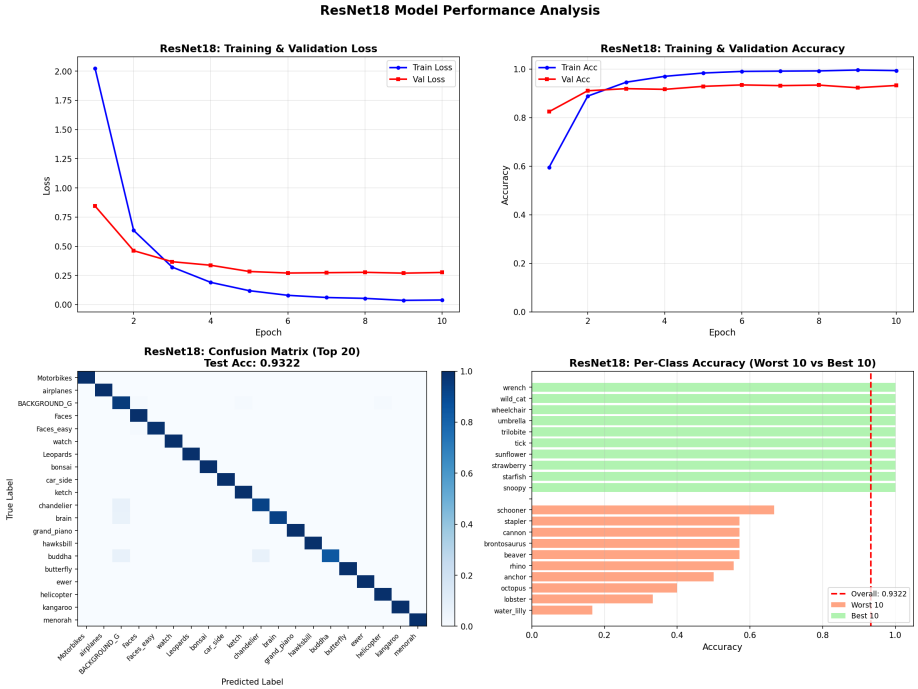

Figure 4: ResNet18 performance: (top) training and validation loss/accuracy curves, (bottom left) confusion matrix for top 20 classes, (bottom right) per-class accuracy.

4 Discussion

4.1 Three Model Comparison

The large performance gap between deep learning models and HOG + SVM highlights the advantage of learned feature representations. HOG captures only local gradient information and, combined with a linear SVM, cannot model complex non-linear decision boundaries. Its sensitivity to class imbalance is reflected in 0% accuracy for several underrepresented classes. In contrast, deep learning models generalize better to categories with limited samples.

EfficientNet-B0 outperformed ResNet18 by 1.75 percentage points despite having fewer parameters (4.1M vs. 11.2M), demonstrating greater architectural efficiency. Its compound scaling and MBConv design allow strong representational capacity with fewer parameters. Both deep learning models achieved top-5 accuracy above 99%, indicating robust class ranking performance.

Class imbalance remains challenging, especially for categories with very few test samples (e.g., lobster with 6 samples). However, EfficientNet-B0 still achieved non-zero accuracy on difficult classes, while HOG + SVM failed completely on several of them, further showing the robustness gained from pre-

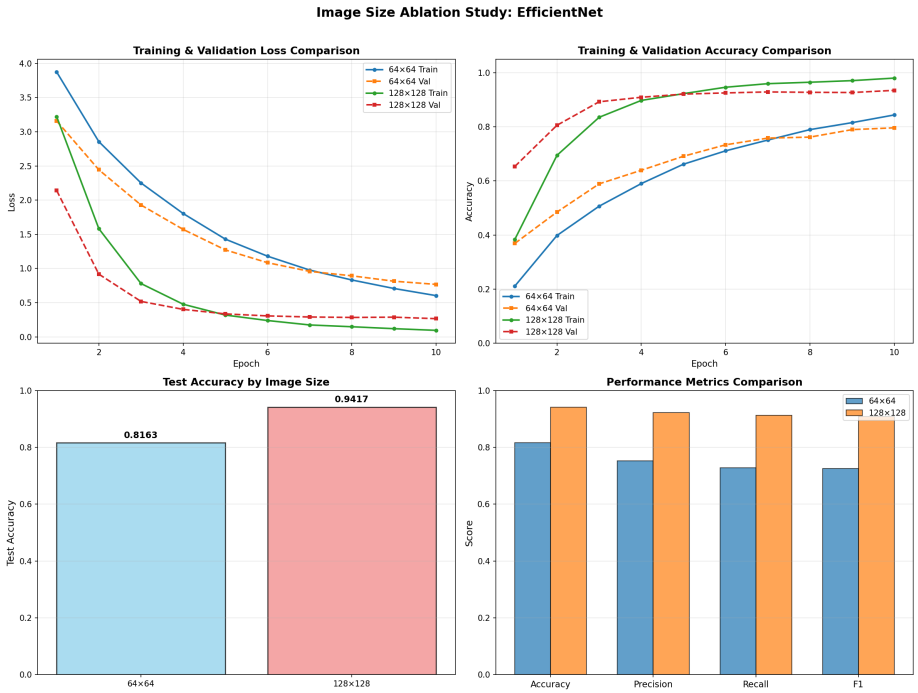

Figure 5: Image size ablation study: (top) training and validation curves comparison, (bottom left) test accuracy comparison, (bottom right) comprehensive metrics comparison.

trained representations.

4.2 Ablation Studies

Increasing input resolution from 64×64 to 128×128 improved accuracy by 12.54 percentage points (81.63% to 94.17%), suggesting that higher resolution preserves fine-grained details important for classification, though at higher computational cost.

Interestingly, training without augmentation slightly outperformed training with augmentation (96.57% vs. 94.97%). This may be because Caltech-101 already contains sufficient variation, and additional transformations can distort discriminative features. In transfer learning settings, pretrained models may already provide strong invariances, so augmentation does not always lead to further gains. This result highlights the importance of empirically validating augmentation strategies rather than assuming they are universally beneficial.

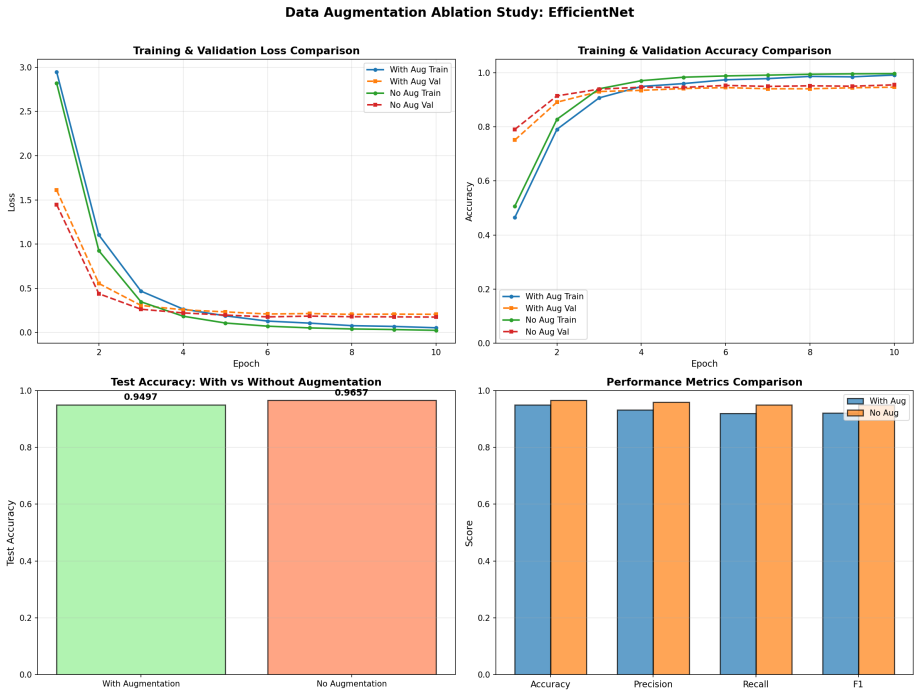

Figure 6: Data augmentation ablation study: (top) training and validation curves comparison, (bottom left) test accuracy comparison, (bottom right) comprehensive metrics comparison.

5 Conclusion

This project evaluated three image classification approaches on Caltech-101: HOG + SVM, EfficientNet-B0, and ResNet18. Deep learning models significantly outperformed the classical baseline, with EfficientNet-B0 achieving 94.97% accuracy compared to 66.47% for HOG + SVM, highlighting the advantage of learned feature representations and architectural efficiency.

Ablation studies showed that increasing image resolution from 64×64 to 128×128 improved accuracy by 12.54%, while training without augmentation slightly outperformed training with augmentation. These results suggest that resolution and augmentation strategies should be evaluated carefully for each dataset.

In practice, EfficientNet-B0 with 224×224 inputs and no augmentation achieved the best performance (96.57%), while 128×128 inputs provide a strong trade-off between accuracy (94.17%) and computational cost. The consistently high top-5 accuracy (>99%) further demonstrates the robustness of the deep learning models.

References

[1] L. Fei-Fei, R. Fergus, and P. Perona, "Learning generative visual models from few training examples: An incremental bayesian approach tested on 101 object categories," *Computer vision and pattern recognition workshop*, 2004, Caltech-101 dataset.

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